

student immersion program
@ NIP (VIP + CX team)

july 8 - 19



Setting up the spectrometer.

DAY 1
July 8, 2024

In the morning (9am to 12nn) we had briefings, and each grad student showed us their research. We needed to sign up for which student to shadow for tomorrow. We learned how to use a GoPro for future activities. So far, I'm very interested in the marine science projects. Lunch was good (Imas also caters here).



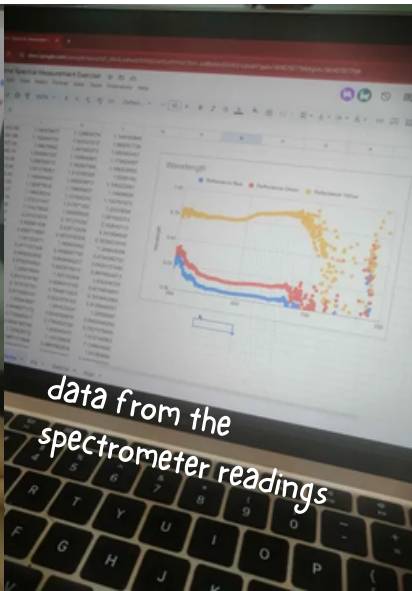
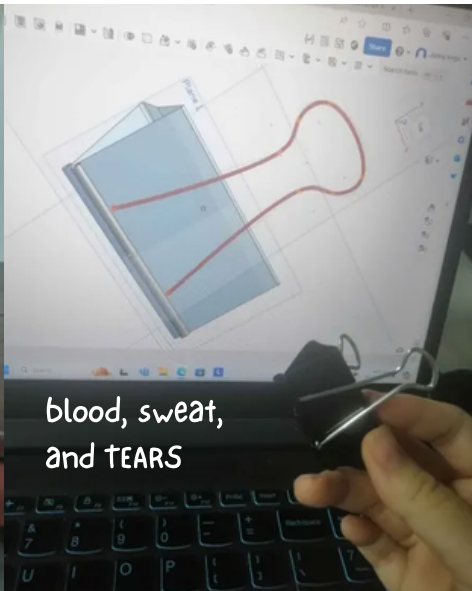
A GoPro Hero 3 from 2014.



After lunch we learned about different types and properties of light and how to use a spectrometer. Ma'am Jing had us guess what color the spectrometer was pointing at just based on wavelengths. After that we learned how to tie several knots (for fieldwork).

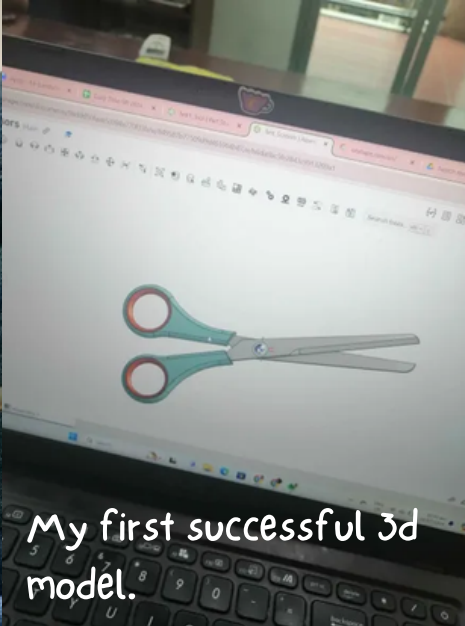
Day 2 July 9, 2024

Ma'am Jing wasn't here today so we learned from her students. Kuya Sean first gave us the basics of Python and how to generate images with it (not turtle.....). Then Ate Andre taught us how to access the world map (by the world bank) about nighttime lights. She's part of the mapping team of VIP which analyzes data remotely (no fieldwork). Then Ate JC showed us how to use a spectrometer to gather actual data, and we used that to calculate for reflectance of a certain color. We also generated a graph for it. It's kind of noisy (the scatterplot) because we weren't in pitch black when recording, there was ambient light so the data got a little messed up. We ate lunch at NIGS and a cat stole our food help. Then the WHOLE afternoon was spent struggling to use OnShape to 3D model a certain office item. I chose a binder clip because it looked easy (it was not). :p





My breakfast.



My first successful 3d model.



Finding eddys.



My fire setup.

Day 3 July 10, 2024

Today it was coding so I was at NIP by SEVEN. I just ate breakfast and then at 8 went to the lab. Kayl and Gab were there. We just did 3D modeling the entire morning (I finished the binder clip, sculpted and animated scissors, and started on my next model). We ate lunch at CS lib, I ate mac&cheese and three (3) meatballs.

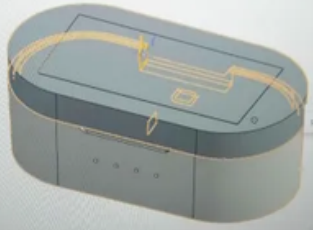
IT WAS SO EMBARRASSING I ALMOST FORGOT TO PAY. But anyway

in the afternoon Ma'am Jing was there and she made us replicate the thesis of one of her students to identify where "eddies" form which are basically vortexes of water where trash and fish congregate so both environmentalists and fishermen pay premium for the information. I was breathing in pepper for AGES because we needed to repeat it three times HAHAAH. I still need to stitch the images together using StarStax (mostly made for star trails) so bye rrrfregrsufeb!!



Where is this?

My second successful 3d model.



Cat in the canteen.



Day 4 July 11, 2024

Today at 9am a guy from Japan (but Filipino) came to the lab and talked about art restoration and preservation especially of cultural heritage pieces, which is what Kuya Jem is doing for his thesis. They use AI and train it based on a certain before (dirty) and after (clean) picture. They only need to clean a small portion manually before the AI can do everything else automatically. The guy who came does scanning mostly and not really restoration, more on preservation. He spent the entire morning talking about all the museums and places around the world he's been to since like 2010 and the scanners he developed, and one of them is NIJI, the one used at this lab (VIP). After that we ate lunch at the NIP canteen (Imas again haha) and then we redid the eddy experiment we did yesterday, except now with a stronger fan and actually floating peppers. Below are the results from both Amihan and Habagat winds. I tried to get the frame-by-frame myself but VLC hates me so Ate Aly did it. One student before already wrote a thesis on Amihan winds, but we're the first to do Habagat. After that I did more 3d modeling on OnShape. I finished modeling my Bluetooth earphone case! :o

Failed attempt because VLC sucks!

Amihan.

Amihan (1 min 22 secs)

Habagat.

Habagat (1 min 35 secs)

Day 5 July 12, 2024

My last day at the VIP group! It's Manansala paintings test measurements day. Ma'am Jing had us text out the setup outside the lab in the morning. I actually ran into Kuya Benji while doing that because he had some training to do at NIP. For the setup, the optical glass wire needs to be steady, so we attached it to a stick secured to a tripod using zip ties. We used a painting to test if the setup was getting the right reflection. Then for lunch (early at 11:00), we ate at the University Hotel and there was SO MUCH food. I got too full hahhsadgjdvsj. We took a jeep to get there after loading Ma'am Jing's car (or truck, it's actually HUGE) and we also walked to the church. There we needed to set up everything and literally measure EVERY COLOR (at least the main ones) using our setup. It took about five whole hours omggg. There was a really pretty cat keeping me company though <3 We met a bunch of people like the owners of the church and the curator of the Philippine Museum. We took a jeep back to the lab and there was free ice cream! I also got to take home some leftover food from the university hotel. I'm so sad to be leaving! Bye VIP :/

Our test setup.



Last photos in the lab while eating ice cream!



The walk <3



Kuya Gab and I killing our arms.



CUTIE!!!



Candid HAHHAHA.

week 1 insights & lessons

I'm fairly certain this is what I want to do in college. All the OJTs there looked like they were enjoying themselves, and this was my first time trying actual oceanography! However, in one week I only scratched the surface of what fieldwork is, especially in the physics aspect. I noticed that there was very minimal computation and the only general physics topic applied was about light and waves. I know there's a lot more to learn but this is the life I'm going to strive for the future. The older students at the lab told us that SIP is meant to build skills and collect as many tools as possible that are practical and have an application in fieldwork since this lab is very heavy on that. Along with that, I also took this as an opportunity to figure out possible research topics. I'm almost grade 10 and that's one of the things expected already by then, and I already have lots of ideas. Ma'am Jing said I'm welcome to drop by the lab any time and join in on the experiments, so maybe I could replicate the eddies experiment but for the Habagat winds, or even in a different location of the world. I wanted to learn more about corals and ocean mapping but unfortunately there wasn't enough time. I enjoyed a lot and I will definitely continue stopping by when I can.

Day 6
July 15, 2024



cable management
final boss.



the main room
of the lab.



the CX team server.



Crimping.



Finally.

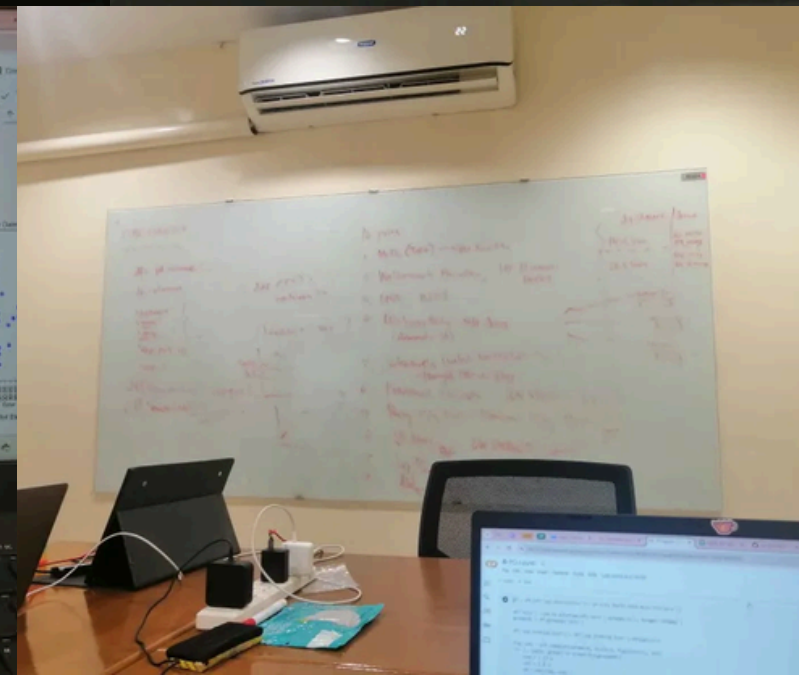
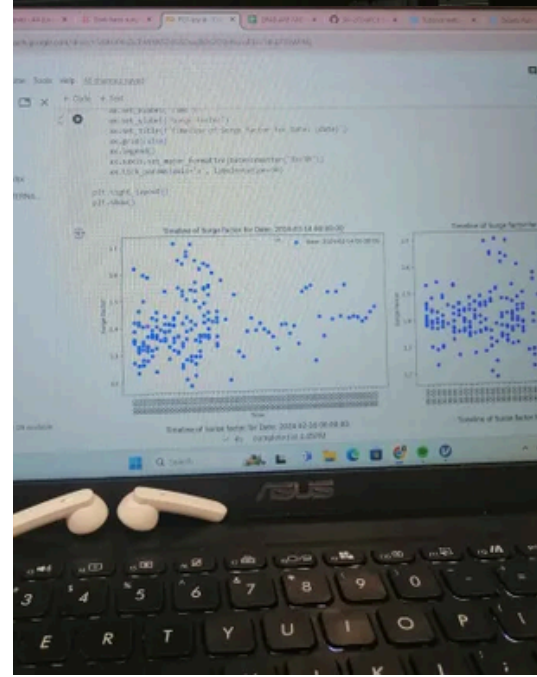
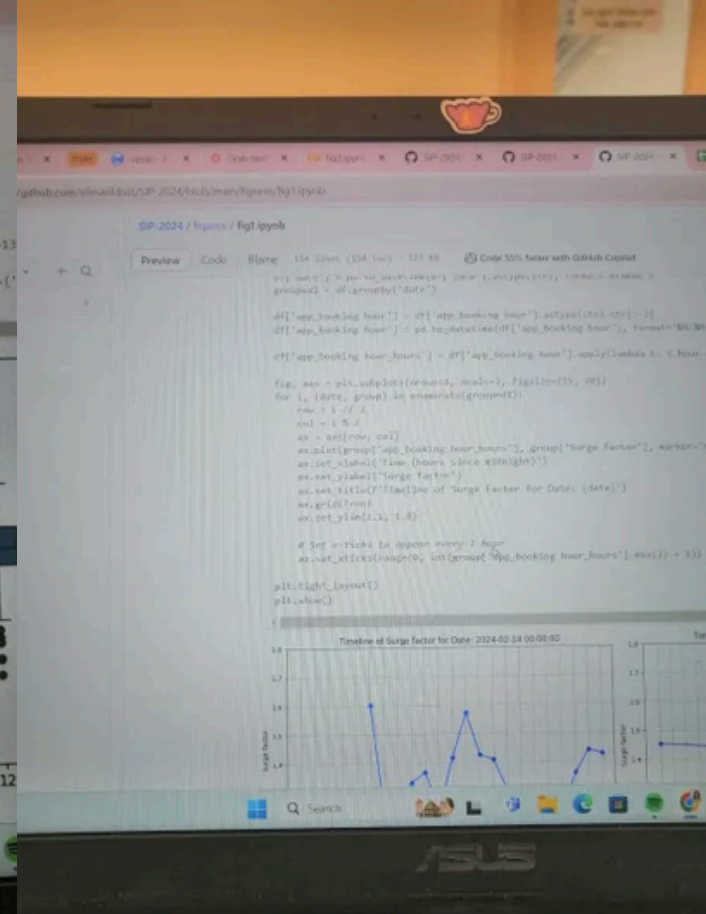
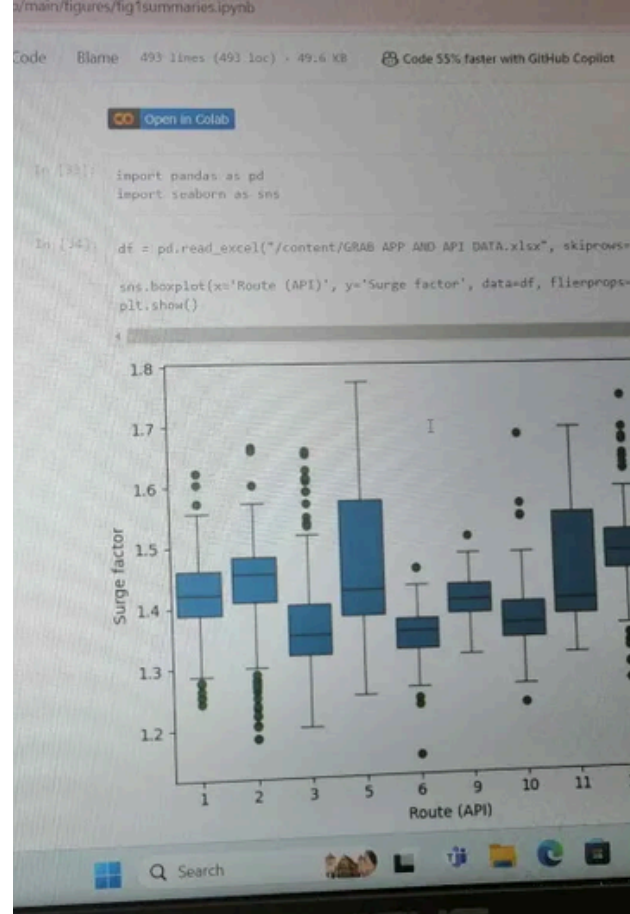
My first day with the Complex Systems team! All the other Pisay students doing their SIP at NIP also started today, so my dad had to show them around. I just went with Dr. Lim to the lab and it is SO COLD and so clean. She said they "religiously mask" meaning I have to smell my own bad breath for eight hours a day (jk). It's a really nice lab though. I noticed that they monitor ventilation and CO2 levels very closely in every room. I'll have to ask them why. The first thing they had us (there were two college OJTs there also, they need 240 hours) do was cable management for the giant server in one room. Then I ate lunch like normal (but I ran into Kuya Gab and Kayl so I ate with them) and in the afternoon we spent like 4 whole hours crimping because I kept messing up :(that basically means cutting wires over and over again it hurt my hands so bad LMFAOISBHB. Dr. Lim told me to learn how to use GitHub, Colab (which I'm already slightly familiar with), and Quarto (which I'm still not sure what it's for) to make a report about what I can learn about Grab fare rates. This will probably take the whole week HAHAAHA. The lab is really nice, very clean, and it's very comfortable too. They have ping pong, a puzzle, and a guitar.

DAY 7 July 16, 2024

I coded the entire day today. I made a GitHub repository and started writing. There are several graphs I'm going to do to create a narrative or gain some context to what I'm doing here. I worked on the same graph the **WHOLE DAY** because there were a lot of things I had to tweak. I'm still not done, but I did learn a lot about how to use pandas and Python. I've also had a little help from AI!

DAY 8 July 17, 2024

I finished the graph! So now I have to do five more haha... It's okay, but now I know how to use GitHub with Colab. I think this will all be useful in the future when I write actual papers. I'm actually just sitting at this chair for almost six hours a day so maybe this is what it feels like to be in academia. Tomorrow I'm going to continue everything I didn't finish today!



Day 9 July 18, 2024

Today was a chill day. I was almost done with my output and I was just cleaning up my GitHub repository. I may have gotten a bit distracted because the people at the lab decided to play hearts, a card game, and I lost miserably. We also played ping pong, and I found out that two of the Masters students at the lab are from Pisay. For my work, I just generated more graphs and started writing down a narrative. It was very reminiscent of my recently-concluded statistics project from last school year. Dr. Lim played ping pong with me before giving comments and suggestions on my writing. I hope I can try publishing it to QuartoPub soon. It wouldn't be an official publication, but it would be nice to know how. And maybe I could make a paper out of this in the future, since the article written by PCIJ and reposted by networks like Rappler were severely lacking in insight (according to Dr. Lim).

Day 10 July 19, 2024

I ate lunch today at UPTC with the GANAP group. While I was there, I also bought donuts from J.co for the lab. Turns out Dr. Lim also ordered pizza using Grab (ironically, the same "opaque algorithm" that my research was on was used for this order as the surge factor was incredibly high). I don't really have any photos even though we did take some, because they weren't sent to me :(. The bulk of this day was me writing narratives and arranging my repository in a coherent way that was up to Dr. Lim's standards. She asked me to send it to her so I'll have to do that soon. My time at this lab also seemed incredibly short, and I wish I could've done two weeks at each lab. Unfortunately, I had plans to travel the day after this, so I had to leave. The people at the CX team were so friendly. One of the kuyas even gave me Pokemon stickers he drew!

week 2 insights & lessons

During my SIP, I got to experience two very different labs. One was full of innovating and fieldwork and the other with coding and AI. I learned so much about the different applications of AI and I'm grateful to Dr. Lim for making me take initiative on the types of questions I want to ask. Although I spent the entire week working on the same thing, it was a glimpse into the real life of a college/grad student. You need to be committed and work on a project that interests you for five years or even more. I was mostly seated at my desk this week, but everything I did was purposeful and I really felt like I was doing something important. Later on, Dr. Lim even praised my work, saying some of the narratives and visuals I created were better than the ones in the official report. This is a career that I also see for my future, but I do want to incorporate some fieldwork because I like to be active. Overall, everything was very enjoyable, everyone was friendly and supportive, but still I got to experience the true college student experience even if only for a week. I even went through a lab's equivalent of "character building" (crimping) which indeed did make me frustrated but did build my character nonetheless. So grateful to everyone at NIP!